PANDASIA
PREVENTING FUTURE PANDEMICS

Communication Activity Report – Year 2024

A transdisciplinary research project funded by the EU and the UKRI

UK Research and Innovation

Pandemic literacy and viral zoonotic spillover risk at the frontline of disease emergence in Southeast Asia to improve pandemic preparedness

PANDASIA

Communication Activity Report – Year 2024

Deliverable: D6.4

Date: December 30, 2024

Dissemination Level: PU-Public document

www.pandasia-project.com

Table of Contents

List of Figures

Project Partners

No.	Abbreviations	Full name	Country	Main roles
1	NMBU	Norwegian University of Life Sciences	Norway	Coordinator WP7 lead
2	NVI	Norwegian Veterinary Institute	Norway	WP4
3	UKHD	Universitätsklinikum Heidelberg University	Germany	WP5 lead
4	IZW	Leibniz Institute for Zoo and Wildlife Research	Germany	WP6
5	QMUL	Queen Mary University of London	UK	WP1
6	CU	Chulalongkorn University	Thailand	WP1 lead
7	UMU	Umeå University	Sweden	WP4 lead
8	KKU	Khon Kaen University	Thailand	WP3 lead
9	MU	Mahidol University	Thailand	WP2 lead
10	SUPA71	SUPA71 Co., Ltd.	Thailand	WP6 lead
11	NOVA IMS	Information Management and Data Science School of the Nova University of Lisbon	Portugal	WP9 Lead

Disclaimer:

1. This project has received funding from the European Union's Horizon Europe research and innovation programme under [Grant Agreement No.101095444], and by UK Research and Innovation (UKRI) under the UK government's Horizon Europe funding guarantee [grant number 10055567].
2. The material presented and views expressed here are the responsibility of the author(s) only.
3. The EU Commission takes no responsibility for any use made of the information set out.

Authors and Acknowledgements

This report was prepared and written by the WP6 project team, led by SUPA71 Co., Ltd (SUPA71), which includes the following members: Dr. Kanokwan Suwannarong, Dr. Cecile M. Lantican, Ms. Kannika Thammasutti, and SUPA71's communication team and technical editor. The team was further supported by Ms. Cora Knoblauch and from the Leibniz Institute for Zoo and Wildlife Research (IZW) team.

Additionally, we would like to express our special thanks to all PANDASIA consortium partners, including the Norwegian University of Life Sciences (NMBU), Norwegian Veterinary Institute (NVI), Universitätsklinikum Heidelberg University (UKHD), Queen Mary University of London (QMUL), Umeå University (UMU), Chulalongkorn University (CU), Khon Kaen University (KKU), Mahidol University (MU), the Information Management and Data Science School of the Nova University of Lisbon (NOVA IMS), and Prof. Dr. Pratap Singhasivanon, for their valuable contributions and collaboration during this reporting process. Additionally, we would like to express our sincere gratitude to Dr. Hans J. Overgaard and Mr. Mohamed Elsayed Abdelgawad Shelil for their insightful suggestions for the Communication Activity Report in the Year 2024.

List of Acronyms

AHV	Animal Health Volunteer
CU	Chulalongkorn University
CSO	Civil Society Organization
CDE	Communication, Dissemination, and Exploitation
DLD	Department of Livestock Development
DMP	Data Management Plan
DOE	Division of Epidemiology
DDC	Department of Disease Control
DHO	District Health Office
DNP	Department of National Parks, Wildlife, and Plant Conservation
DRU	Dhonburi Rajabhat University
ELISA	Enzyme-Linked Immunosorbent Assay
ERB	Ethics Review Board
EU	European Union
EPI-WIN	WHO's Epidemic Intelligence from Open Sources
FAIR	Findable, Accessible, Interoperable, and Reusable
FGD	Focus Group Discussion
GLEWS	Global Early Warning and Prevention System
HPAI	Highly Pathogenic Avian Influenza
IACUC	Institutional Animal Care and Use Committee
IZW	Leibniz Institute for Zoo and Wildlife Research
KKU	Khon Kaen University
KII	Key Informant Interview
MFU	Mae Fah Luang University
MOPH	Ministry of Public Health
MNRE	Ministry of Natural Resources and Environment
MOAC	Ministry of Agriculture and Cooperatives
MU	Mahidol University

MUNA	Mahidol University, Nakhon Sawan Campus
NMBU	Norwegian University of Life Sciences
NORDITA	The Nordic Institute for Theoretical Physics
NOVA IMS	Information Management and Data Science School, Nova University of Lisbon
NVI	Norwegian Veterinary Institute
OIC	Office of International Cooperation
ONEP	Office of Natural Resources and Environmental Policy and Planning
OSF	Open Science Forum
PHOs	Provincial Health Offices
PI	Principal Investigator
POC	Point-of-Care
PRs	Press Releases
QMUL	Queen Mary University of London
SAB	Scientific Advisory Board
SCM	Steering Committee Meeting
SCPHT	Sirindhorn College of Public Health Trang
SEAMEO TROPMED	Southeast Asian Ministers of Education Organization Tropical Medicine Network
SUPA71	SUPA71 Co., Ltd.
SYGMA	European Commission's System for Grant Management
TALENOME	Talenome DNA Professional Co., Ltd.
UKHD	Universitätsklinikum Heidelberg University
UMU	Umeå University
VHV	Village Health Volunteer
WP	Work Package
ZPO	Zoological Park Organization

1. Executive Summary

PANDASIA is a transdisciplinary research project aiming to understand the complexity of zoonotic spillover, thereby enhancing global pandemic preparedness. PANDASIA adopts the concept of "One Health (OH)" to investigate potential pandemic drivers along nature-rural-urban gradients through the comprehensive collection of social, biological, and molecular data combined with predictive modeling. By integrating diverse perspectives, PANDASIA aims to develop evidence-based strategies for mitigating zoonotic risks and strengthening public health systems at local, national, and global levels.

Work Package 6 (WP6) facilitates the **exploitation** of project results to **disseminate and communicate** the project's results and translate these into usable formats to be accessed by targeted audiences such as local communities, local government units (provinces, cities, municipalities, villages), the Thai government, Thai citizens, regional authorities and organizations in Southeast Asia, as well as global entities such as EU and international stakeholders, international development agencies, media representatives, academia, scientific institutions, the scientific community, civil society organizations, and consortium partners. In this regard, WP6 prepared and updated the **Knowledge Exchange Strategy Report** regularly based on the actual situation and feedback from various relevant stakeholders, which updates the overall communication goal of PANDASIA WP6 for implementation from 2023 to 2027. Among the main tasks of WP6 support each PANDASIA consortium partners to promote the project's activities, outputs, and results to stakeholders in Thailand and the EU.

On June 9, 2023, the **PANDASIA website** was launched (https://pandasia-project.com/). In addition, WP6 made use of other social media platforms, like Facebook, LinkedIn, X (formerly Twitter), and YouTube, to share the project's activities, outbreak news, and project implementation results. The WP6 team also enhanced the visibility of the project among Thai stakeholders by distributing information packages to participants during workshops/meetings, consisting of tote bags, T-shirts, caps, and pens that bear the PANDASIA branding to visualize the project. From 2023 to December 24, 2024, the PANDASIA website and social media platforms (Facebook, LinkedIn, X, and YouTube) were actively maintained and updated, achieving significant engagement globally, with 21,588 website page views and thousands of interactions across platforms. Across all social media platforms received a grand total of 8,653 followers and connections. A total of 43 outbreak news updates, 12 activity updates, and 30 event posts were shared to engage audiences. Communication materials, including press releases, newsletters, and flyers in English and Thai, were downloaded 495 times.

The WP6 led by SUPA71 Co., Ltd (SUPA71) with support from the Leibniz Institute for Zoo and Wildlife Research (IZW) documented project activities and produced videos to capture the support and knowledge-sharing activities among local partners. The video materials were edited to protect the personal privacy of the study participants. By signing the informed consent form developed by PANDASIA WP6, they expressed their willingness to participate in the dialogue and be recorded as part of the event. Additionally. newsletters and videos of the workshop activities were produced periodically. These materials were uploaded to the website in January 2024 after obtaining approval from the PANDASIA Principal Investigator (PI). WP6 also established internal communication guidelines for consortium members to ensure ethical and coordinated outreach.

Moreover, the WP6 team started to connect with the press offices of consortium members. A roster of media contacts was initially developed by the Leibniz Institute for Zoo and Wildlife Research (IZW). To date, IZW and SUPA71 have maintained communication with the press offices of the consortium members. A media contact list was developed to improve outreach across Europe, Thailand, and Southeast Asia to promote PANDASIA's activities and results. In 2023-2024, WP6 collaborated with other WPs to gather detailed updates with results, as follows:

- **Work Package 1 (WP1): Society, Livelihoods, and Pandemic Risk Perceptions,** led by Chulalongkorn University (CU), in collaboration with Heidelberg University (UKHD) and Queen Mary University of London (QMUL) identified and meet potential stakeholders, as well as obtained necessary ethical and local approvals. WP1 conducted stakeholder mapping, coproduction workshops, and qualitative baseline data collection in Chanthaburi and Chiang Rai to identify human and societal factors contributing to zoonotic spillover risks.
- **Work Package 2 (WP2): Wildlife, Livestock, and Land-Use Change,** led by Mahidol University (MU), WP2 focused on identifying animal hosts, land-use changes, and spillover interfaces. The WP2 team started collecting data on rodents, bats, livestock, and environmental samples to build predictive models.
- **Work Package 3 (WP3): Virology**, led by Khon Kaen University (KKU), established protocols for sample collection and advanced viral identification techniques. It aimed to identify viruses with zoonotic potential, laying the groundwork for developing a point-of-care test (POCT).
- **Work Package 4 (WP4): Mathematical Modeling of Spillover Risks**, led by Umeå University (UMU), developed ecological and epidemiological models to analyze spillover processes. The models were designed to integrate data from other WPs, providing insights into spillover dynamics and potential mitigation strategies.
- **Work Package 5 (WP5): Pandemic Preparedness Literacy**, led by Universitätsklinikum Heidelberg University (UKHD), focused on developing a community-based pandemic literacy intervention. It designed a baseline questionnaire and conducted systematic reviews to inform its approach, benefiting other WPs as well.
- **Work Package 6 (WP6): Dissemination, Exploitation, and Communication**, led by SUPA71 Co., Ltd (SUPA71) in collaboration with Leibniz Institute for Zoo and Wildlife Research (IZW) supported the dissemination and communication of project activities, enhanced the knowledge-sharing strategies by profiling stakeholders and engaging

media contacts. Collaborations with other WPs ensure effective outreach to global and regional audiences.

- **Work Package 7 (WP7): Coordination and Management**, led by Norwegian University of Life Sciences (NMBU) facilitated the coordination among partners, maintaining the project's timeline and objectives.
- **Work Package 8 (WP8): Ethics**, led by NMBU in collaboration with the external committee to oversee ethical compliance across all research activities ensured adherence to ethical standards in human and animal studies.
- **Work Package 9 (WP9): One Health Interventions and Community Engagement,** led by the Information Management and Data Science School of the Nova University of Lisbon (NOVA IMS) prepared community-driven approaches for the One Health interventions, focusing on community engagement activities for pandemic risk reduction.

2. Introduction

PANDASIA is a transdisciplinary research project aiming to understand the complexity of zoonotic spillover, thereby enhancing global pandemic preparedness. This will be done through a comprehensive collection of social, biological, and molecular data with predictive modelling of zoonotic spillover rates and disease emergence in high-risk settings in Thailand. This interdisciplinary approach will enable an intensive understanding of potential pandemic drivers along nature-rural-urban gradients. PANDASIA consists of nine work packages (WPs), as shown in Figure 1.

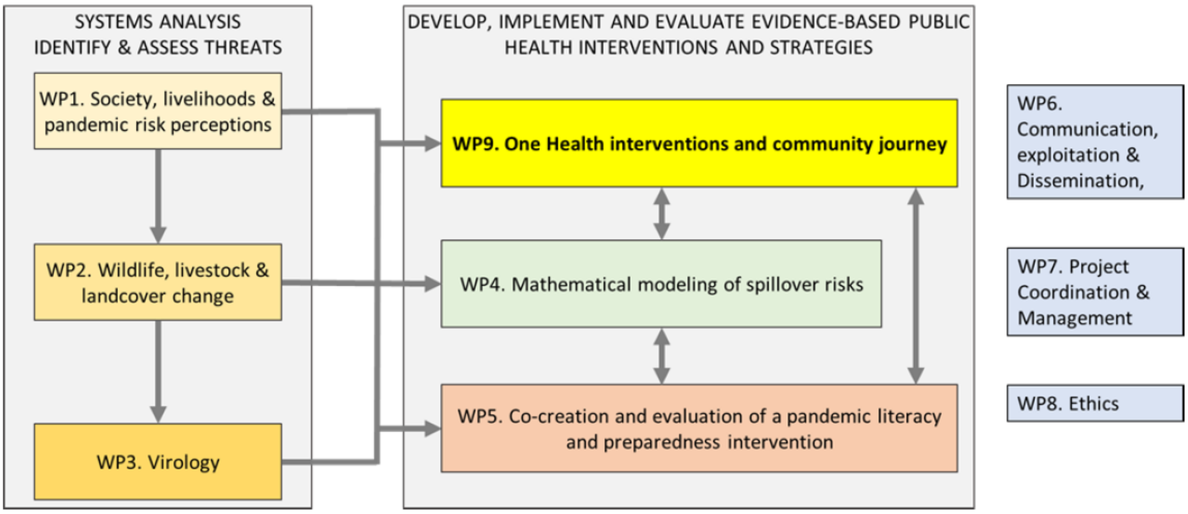

Figure 1. *PANDASIA Work Packages*

PANDASIA has the following Specific Objectives (SO):

SO1. Determine and monitor human and societal factors impacting zoonotic spillover risk by identifying populations, human behaviours, human-animal-environmental interactions, structural drivers and barriers, and relevant policies in study locations (WP1& WP9).

> **RR1.** Human and societal factors of importance for zoonotic spillover risk that may impact human, animal, and environmental health identified.
>
> **RR2.** Key policies, policy actors, and stakeholders relevant to zoonotic spillover risk prevention and control measures at local and national levels identified.
>
> **RR3.** Policymakers, relevant government sectors, local authorities, and communities engaged to obtain feedback and recommendations for the communication interventions.

SO2. Determine the most important wild and domestic vertebrate animal hosts of importance for spillover in each study location, assessing their presence, abundance, and diversity (WP2).

> **RR4.** Presence and population characteristics of wild vertebrate and domestic animal hosts.

SO3. Assess and model the potential effect of changes in land use, land cover, climate, and human demographic factors since the year 2000 on spillover risk since 2000 (WP2).

RR5. Changes in historical land use, land cover, climate, and human population that may affect spillover events determined and modelled.

SO4. Identify previously unrecognized pathogens with spillover potential, specifically targeting viral groups in vertebrate animal hosts for which there is strong precedence for viral occurrence and emergence in the region (WP3).

RR6. Ranked list of viruses with spillover potential with high binding affinity to host cell receptors and infection capacity based on in silico and wet lab-based criteria.

RR7. Viruses with spillover potential identified and characterised in specific animal species and study locations.

RR8. Identified viruses serving as empirical data to generate highly accurate risk models.

RR9. Identified viruses used as targets in Point of Care Test development.

SO5. Develop ecological, epidemiological, and evolutionary conceptual models to enhance understanding of the significance of the general principles and pathways of spillover processes and to guide the development and analyses of a structurally equivalent, yet more specific, data-driven model (WP4).

RR9. A flexible epidemiological network model developed that captures essential steps of the spillover process.

RR10. An ecological conceptual model developed that couples epidemiology and virulence evolution.

RR11. Evolutionary and epidemiological frameworks integrated in a spatially explicit model relevant to study areas.

SO6. Develop a point-of-care (POC) virus test kit prototype for healthcare providers and professionals to identify spillover at the earliest stages to prevent epidemic or pandemic spread (WP3).

RR12. ELISA protocol developed and used as a standard method to detect and identify spillover viruses.

RR13. Prototype POC developed with high sensitivity and specificity compared to ELISA

SO7. Create and test a critical public health measure – a pandemic prevention and preparedness literacy (3PL) intervention to reduce zoonotic transmission and pandemic risk (WP5& WP9).

RR14. A conception (published as a policy brief and academic paper) of pandemic prevention and preparedness literacy.

RR15. A co-created 3PL intervention (tested in a quasi-experimental causal impact evaluation, with an associated protocol and primary results papers and a policy brief and toolkit for implementation in other communities).

In addition, PANDASIA-WP6 implements the PANDASIA's **Knowledge Sharing Strategy**. The strategy has a three-pronged approach – **communication, dissemination, and exploitation (CDE)**. We define **communication** as sharing information, data, and knowledge. This could be done internally between the consortium members, Scientific Advisory Board, and Ethics Review Board or externally in all kinds of interactions with stakeholders. We also define **dissemination** as sharing any output from our consortium with specific guidance, ideas, publications, policy papers, and algorithms. We engage with various stakeholders to apply, test out, and develop user-friendly tools, apps, policy briefs, and technical guidance to achieve this task. On **exploitation**, we ensure the sustainability of our work beyond the project's lifespan. We aim to establish sustainable partnerships with various academic institutions, CSOs, and international and non-governmental organizations. We aspire to utilize our results to generate national and regional policy guidance to enhance national and regional application of the International Health Regulations (2005) and Pandemic prevention policies.

2.1 Objectives

PANDASIA's communication activities seek to facilitate dialogue between science and society. Through these activities, it discloses the overall project goals, planned contributions of the participating organizations/ the consortium partners, and their activities, given their multidisciplinary expertise. However, to ensure that project goals and planned actions and interventions will be successfully achieved, WP6 closely coordinates with other WPs that provide the content to be delivered to target audiences.

3. Relevant Activities in 2024

3.1 Relevant Activities from PANDASIA Consortium Partners

Work Package 1 (WP1)

Work Package 1 (WP1), led by Chulalongkorn University (CU), collaborated with the University of Heidelberg (UKHD) and Queen Mary University of London (QMUL), made significant progress on several key activities towards meeting the objectives of identifying and monitoring human and societal factors affecting zoonotic spillover risks, focusing on populations, human behaviors, human-animal-environment interactions, structural drivers and barriers, and relevant policies in the study locations. The activities conducted under WP1 in 2023-2024, are as follows:

Scoping visits, permissions for PANDASIA activities in study locations, and systematic reviews: From March to October 2023, the CU team worked with government agencies and local authorities in Thailand to gain support and permissions for the PANDASIA project. Meetings were held with the Ministry of Public Health (MOPH), the One Health National Committee, Provincial Health Offices (PHOs), District Health Offices (DHOs), and Provincial Livestock Offices in Chanthaburi and Chiang Rai, as well as the Department of Livestock Development (DLD) under the Ministry of Agriculture and Cooperatives (MOAC). From March through April 2023, the CU team visited Mae Fah Luang and Wiang Kaen districts in Chiang Rai and Pong Nam Ron and Soi Dao districts in Chanthaburi to introduce the project and secure local approvals. These sites were chosen based on zoonotic disease data, feedback from officials working on human-animal interactions, and the willingness of local authorities to participate. Additionally, permissions for PANDASIA activities were obtained, including from Chiang Rai PHO approval on November 13, 2023, and Chanthaburi PHO approval on July 20, 2023, and livestock offices in both provinces in June 2023. These approvals allowed the CU team to carry out health, livestock, and zoonotic disease prevention activities, which were reported in deliverables D1.1 and D6.3. As of the first 18-month report of PANDASIA, **systematic reviews** by UKHD are in progress. One systematic review will be uploaded to the EU portal (D1.3) and will be submitted for publication in September 2024.

Stakeholder co-production workshops: In October 2023, WP1 team, led by Prof. Dr. Amonsin and Dr. Suwannarong from the Faculty of Veterinary Science, Chulalongkorn University (CU), collaborated with experts including Dr. Bärnighausen from the University of Heidelberg (UKHD) and Dr. Montag from Queen Mary University of London (QMUL), and the team from various universities, such as Khon Kaen University, Sirindhorn College of Public Health Trang (SCPHT), Mae Fah Luang University (MFU), Mahidol University (MU), Nakhon Sawan Campus (MUNA), and Dhonburi Rajabhat University (DRU) to conduct the stakeholder mapping workshops at the MeStyle Museum Hotel in Bangkok, Thailand, on October 17 to 19, 2023 (Figure 2). The workshops included three focus group discussions (FGDs). Two of these FGDs were held with key government officers/ public health and animal authorities from central and provincial levels on 17-18 October 2023. The other FGD was conducted with representatives from CSOs in Thailand on 19 October 2023. The CU team completed the Thai transcriptions of three FGDs and summarized all activities and discussions from the stakeholder workshops.

Figure 2. *PANDASIA Stakeholder Mapping Workshops, October 17 to 19, 2023*

Research ethics approvals: On June 20, 2023, the Chulalongkorn's University Research Ethics Committee for Research Involving Human Participants approved PANDASIA's research proposal (Protocol No. 660079). CU team led the entire process of preparation, documentation, and application requirements. A detailed report on ethics approvals (D1.5) was submitted on June 30, 2023. From June through July 2024, the CU team obtained amendments and renewals of ethical approvals for research involving humans. Amendment approval for human research was obtained from Chulalongkorn University's Research Ethics Committee in June 2024, while renewal ethical approval for human research was in July 2024 [Annex 1]. For research involving animals, the WP1 team obtained ethics approval from the Institutional Animal Care and Use Committee (IACUC) of the Faculty of Veterinary Science, Chulalongkorn University, on May 24, 2023, with a detailed report (D1.5) also submitted on June 30, 2023. Renewal of approval for animal research ethics was also granted by the IACUC in May 2024. These approvals would ensure compliance with ethical standards for research involving both human and animal subjects [Annex 2].

Baseline qualitative data collection conducted in Chanthaburi: From November 8 to 19, 2023, the WP1 team conducted qualitative baseline data collection in four study locations in Chanthaburi province. The CU team completed 32 in-depth interviews (IDIs), 10 focus group discussions (FGDs), and 30 shared walks. The CU team transcribed 32 IDIs and 10 FGDs, as well as provided summaries of 30 shared walks. In addition, CU and UKHD pre-tested the quantitative baseline questionnaire among villagers in the study locations to determine the validity and precision of the data collection tools. In March 2024, CU conducted 10 additional pre-tests, followed by validation and backtranslation of the questionnaire with UKHD. The finalized Thai and English versions were submitted to Chulalongkorn University's IRB, receiving amendment approval on June 5, 2024, and an extension on July 23, 2024. The analysis of these baseline qualitative study will be carried out by UKHD, focusing on key findings related to the village health volunteers (VHVs) and zoonotic spillover risk factors.

Policy analysis and stakeholder mapping data collection in Chanthaburi province, Chiang Rai province, and Central levels: WP1, led by CU with support from QMUL and UKHD conducted policy analysis and stakeholder mapping to gather information from central and provincial government stakeholders, local authorities, and community actors at national, provincial, district, and community levels. CU completed 22 key informant interviews (KIIs) and 5 focus group discussions (FGDs) in Chanthaburi in September 2023 and 18 KIIs and 4 FGDs in Chiang Rai in January 2024 (Figure 3). After data collection, the CU team completed transcriptions of all data collected during stakeholder mapping activities, including 22 KIIs and 5 FGDs in Chanthaburi and 18 KIIs and 4 FGDs in Chiang Rai.

Figure 3*. Stakeholders' mapping data collection in Chanthaburi and Chiang Rai Provinces of Thailand*

For the **policy documents**, CU QMUL, and UKHD developed inclusion and exclusion criteria and a list of potential data sources and institutions. CU team sent official request letters in July 2024 and received national policy documents from eight organizations, including, 1) the Department of Livestock Development (DLD) of Ministry of Agriculture and Cooperatives (MOAC), 2) the Office of Natural Resources and Environmental Policy and Planning (ONEP) of Ministry of Natural Resources and Environment (MNRE), 3) the Department of Water Resources (DWR) of Ministry of Natural Resources and Environment (MNRE), 4) the Coordinating Unit for One Health (One Health), Division of Communicable Diseases, Department of Disease Control (DDC) of

Ministry of Public Health (MOPH), 5) the Office of International Cooperation (OIC), Department of Disease Control (DDC), Ministry of Public Health (MOPH), 6) the International Health Regulation Thailand National Focal Point (IHR Focal Point), Division of Epidemiology (DOE), Department of Disease Control (DDC), Ministry of Public Health (MOPH), 7) the Department of National Parks, Wildlife and Plant Conservation (DNP), and 8) the Zoological Park Organization (ZPO). CU also shared the policy documents to QMUL and UKHD in August 2024, including 15 international policy documents, 18 national policy documents, 11 regional policy documents, 26 provincial policy documents, and 34 websites. These documents were collected to support policy analysis and stakeholder mapping activities.

In addition, the CU team prepared letters for additional KIIs requests at the national levels in July 2024 and coordinated appointments with relevant government officials and civil society representatives at the national level in August 2024 for QMUL and UKHD to conduct interviews. In September to October 2024, QMUL and UKHD conducted seven additional KIIs online, with support from the CU team by arranging Zoom meetings and coordinating with the key informants to ensure smooth communication and scheduling. Also, CU completed the translation and validation of the selected 23 transcriptions, as well as identifying 12 FGD speakers of the stakeholders' mapping data collection in Chanthaburi and Chiang Rai provinces, and workshops, while QMUL validated the accuracy of the translations. The CU team also arranged meetings with QMUL, UKHD, and the PI from the NMBU to facilitate the handover of tasks after October 2024. Between October to December 2024, QMUL and UKHD worked on analyzing and drafting **the policy and stakeholder analysis and mapping report**.

Work Package 2 (WP2)

Work Package 2 (WP2), led by Dr. Siriaroonrat from the Faculty of Environment and Resource Studies, Mahidol University (MU), included co-principal investigators Prof. Dr. Duangkae from the Faculty of Forestry, Kasetsart University; Assoc. Dr. Chaiyes from the School of Agriculture and Cooperatives, Sukhothai Thammathirat Open University; Assoc. Dr. Chaisiri from the Faculty of Tropical Medicine, Mahidol University; Assoc. Dr. Sripiboon from the Faculty of Veterinary Medicine, Kasetsart University; Dr. Tipkantha and Dr. Jairak from the Zoological Park Organization; and veterinary epidemiologist Lekcharoen.

According to the first 18-month report of PANDASIA, WP2 focused on studying wild and domestic animal hosts and preparing for fieldwork in 2023. The WP2 prepared ethics approval for wild animal research in 2023 and obtained the approval in March 2024. During 2023, WP2 reviewed previous studies on mammals in Chanthaburi and Chiang Rai provinces, collected land use data, and conducted site visits to select target species and sampling locations. Collaboration with local authorities ensured smooth coordination of field activities. WP2 also compiled historical data on land use, land cover, and climate to better understand factors influencing zoonotic spillover risks. Plans were established to continue sampling in the wet season of 2024 and to analyze collected samples for virus identification and host associations.

These activities demonstrate significant progress in understanding zoonotic spillover dynamics and the environmental and ecological factors involved.

In 2024, WP2 conducted biological sampling and ecological data collection from bats, rodents, secondary wildlife, and domestic animals during the dry and wet seasons in Chanthaburi and Chiang Rai. Wildlife samples were collected from diverse habitats, including forests, agricultural areas, dump sites, and households.

During the dry season, samples were collected in Pong Nam Ron and Soi Dao districts, Chanthaburi, from April 2–8, and in Wiang Kaen and Mae Fah Luang districts, Chiang Rai, from April 21–29. The WP2 team obtained oral swabs, rectal swabs, blood samples, and more from 178 bats, 172 rodents, 161 domestic animals (dogs and cats), and 15 secondary wildlife species (e.g., Tupaia, Menetes, squirrels, and civets) (Figure 4).

Figure 4. *Dry season biological sampling in Chanthaburi and Chiang Rai, April 2024*

In the wet season, the sampling activity took place in Pong Nam Ron and Soi Dao districts, Chanthaburi, from July 8–15, and in Wiang Kaen and Mae Fah Luang districts, Chiang Rai, from October 18–26. The WP2 team collected similar samples from 234 bats, 132 rodents, 155 domestic animals (dogs and cats), and 9 secondary wildlife species (e.g., Tupaia, Menetes, and birds). All samples were submitted to the KKU laboratory for virus screening (Figure 5).

Figure 5. *Wet season biological sampling in Chanthaburi and Chiang Rai, July–October 2024*

From August 13–23, 2024, the invertebrate team revisited the Chanthaburi study sites to conduct mosquito, fly, and leech collections during the wet season. The team, led by Dr. Overgaard and Dr. Brown (medical entomologists) from Khon Kaen University and the Norwegian University of Life Sciences (KKU/NMBU), included Ms. Chaiphonngam (logistics officer) and local entomologists Mrs. Penparpa, Ms. Maneerat, Mr. Nanthaphop, and Mr. Pramote from the Government Office of Vector-Borne Disease Control. They were joined by Entomology Field Assistant Mr. Phun and Ms. Viriya, an MSc student from the Faculty of Tropical Medicine, Mahidol University (Figure 6).

Figure 6. *Wet season invertebrate sampling activities in Chanthaburi, August 13–23, 2024*

The invertebrate team spent 10 days collecting invertebrates from the same locations as the dry season fieldwork, focusing on zoonotic spillover interfaces such as village houses, temples, dump sites, fruit orchards, forest edges, interiors, and bat caves. Mosquitoes were collected to assess their role as intermediate hosts in virus transmission, while flies, leeches, and blood-fed mosquitoes were collected for environmental DNA (eDNA) analysis to identify bloodmeal hosts and assess animal biodiversity. The invertebrate sampling was aimed to identify mosquito and animal species involved in transmission and highlight interfaces posing a high risk of virus spillover to humans. A total of 8,439 mosquitoes (37 blood-fed), 1,771 flies, and 2 leeches were collected, identified, and stored for subsequent virus screening at the KKU laboratory.

Work Package 3 (WP3)

In 2023, Work Package 3 (WP3), as detailed in the first 18-month report of PANDASIA, WP3 led by Khon Kaen University (KKU), focused on key activities related to virus identification and spillover risk assessment. The WP3 team worked on processing biotic and abiotic samples collected from fieldwork to ensure high-quality preparation for virome profiling and viral host identification. This included applying advanced methodologies to manage and analyze samples effectively. Additionally, WP3 began developing virome profiles and identifying viral hosts to understand the diversity and potential transmission pathways of viruses. In addition, WP3 worked on preparing methodologies for assessing the risk of zoonotic spillover, which included integrating computational tools and laboratory techniques for future analyses. The activities in 2023 laid the groundwork for ongoing tasks and deliverables critical to understanding viral dynamics and their potential spillover risks.

During the first quarter of 2024, WP3 focused on developing equipment, tools, and methodologies for environmental sample collection, conducting trial collections within Khon Kaen University as a pilot site. This phase laid a solid foundation for subsequent field operations. During this period, primers for PAN-viral PCR were successfully designed, representing a significant step forward in detecting and identifying a wide range of viruses. A strategic planning meeting was held to finalize geographic coordinates for collecting biotic and abiotic samples. Using data from WP2, key locations in Chanthaburi and Chiang Rai provinces were identified and subdivided into five distinct categories: forest, agricultural, dumpsite, villages, and temple. These initial activities ensured a structured and targeted approach to sample collection.

On the second quarter, WP3 introduced a trial version of a dashboard for the viral database, providing streamlined access to information relevant to Southeast Asia. This tool integrates data from the NCBI database and a thorough review of scientific literature, offering critical support for the research team. During this period, WP3 initiated the validation of virus analysis methods to ensure accuracy and reliability in subsequent studies. Field teams conducted on-site environmental sample collections in Chanthaburi and Chiang Rai provinces, collaborating closely with WP2 and the invertebrate team. The samples included various biological specimens from animals and environmental samples such as air, water, and sediment. The selection of target species focused on animals identified by WP2 as having high potential for zoonotic disease transmission. WP3 also collected additional samples of mosquitoes and flies to assess their role in disease dynamics. These efforts were a crucial step in building a comprehensive understanding of viral activity in the region.

During the third quarter, WP3 ramped up preparations for wet season sample collections (Figure 7). Fieldwork in Chanthaburi was successfully conducted in July through August, as planned. However, severe flooding in Thailand led to the postponement of field activities in Chiang Rai to November. Despite these disruptions, the WP3 team pursued the collaboration with partners to setup conditions for PAN-viral qRT-PCR system, and for Prof. Alex Greenwood from the Leibniz Institute for Zoo and Wildlife Research (IZW) to validate virus enrichment and sequencing protocols using the Twist comprehensive viral panel and Illumina sequencing platform.

Figure 7. *WP3*'s *wet season fieldwork and sample collection activities*

The fourth quarter represented the culmination of WP3's efforts throughout the year. WP3 team conducted a hybrid capture training workshop in collaboration with experts from Twist Bioscience (Figure 8). This training equipped the WP3 team with essential skills and techniques for advanced sample preparation and sequencing, ensuring high-quality data for downstream analyses. Field activities in Chiang Rai were also successfully completed in November, overcoming earlier delays. Following the wet season sample collection, the WP3 team transitioned to blood sample collection from targeted populations in Chanthaburi during early December. These activities were aimed to evaluate immune responses and integrate findings into the broader research framework. At once, WP3 concentrated on consolidating its progress, analyzing preliminary data, and refining methodologies for future studies.

Figure 8. *WP3's hybrid capture training workshop and laboratory activities*

During the year, WP3 achieved critical milestones in sample collection, primer design, database development, and method validation. These accomplishments reflect the team's commitment to addressing the complexities of viral research. For 2025, WP3 will focus on laboratory analysis of the collected samples to ensure comprehensive data integration and actionable insights. These efforts would provide valuable information to enhance preparedness and prevention strategies for future emerging infectious diseases, contributing to global health resilience.

Work Package 4 (WP4)

Work Package 4 (WP4), led by Umeå University (UMU), made significant progress in developing mathematical models to assess spillover risks and support pandemic preparedness. In 2023, the WP4 team created the first version of a conceptual eco-evolutionary modeling framework, connecting the natural environment, wild and domestic animals, pathogens, and humans to spillover risk. Additionally, progress was made in identifying ecological and evolutionary drivers of spillover using advanced methodologies such as Bayesian inference and data-driven approaches. These efforts had laid a strong foundation for future work, contributing to the development of tools and models to enhance spillover risk management and pandemic preparedness strategies.

In 2024, WP4 focused on creating and improving models to understand how zoonotic diseases spill over from animals to humans and to support better pandemic preparedness. The WP4 team worked on developing process-based models that consider how populations of wild animals, domestic animals, and humans interact, as well as how diseases spread and evolve. These models were designed to identify areas where spillover risks are highest and to test strategies for reducing those risks.

In February 2024, WP4 hosted a two-day hybrid workshop at Umeå University from February 26-17, bringing together researchers to discuss advancing modeling efforts and strengthening collaboration across work packages. The workshop was aimed to identify key processes and parameters for integration into the models developed in Task 4.1, Task 4.2, and related tasks. It also sought to clarify work and data dependencies for parameterizing and refining models, enhance feedback and collaboration between WP1-4, and align data collection efforts with the needs of data-driven model development (Figure 9).

Figure 9*. WP4's hybrid workshop at Umeå University, February 26-17, 2024*

One of the key tasks in 2024 was improving individual-based models. These models were designed to simulate the daily activities of animals like fruit bats, including where they roost, forage, and travel and how these activities overlap with human behaviors. The goal was to understand how land use and human-animal interactions increase spillover risks. The model also would look at how changes, such as land-use shifts or climate impacts, might influence bat behavior and spillover risks.

The WP4 team also worked on understanding how diseases evolve when they move between species. The WP4 team developed models to study how pathogens adapt to new hosts and what conditions might lead to new strains of diseases. Early findings suggested that the closeness of species and the frequency of contact between them play a big role in how diseases evolve and spread.

Collaboration with other work packages was an important part of WP4's work in 2024. WP4 worked with WP9 to prepare for workshops that will engage experts and use their inputs/insights to create scenarios for pandemic preparedness. The WP4 team also used data collected by other work packages to make their models more accurate and relevant to real-world situations.

Throughout the year, WP4 held regular meetings and workshops to share knowledge and work together with other teams. This collaborative approach helped ensure that the models are practical and can be used to develop tools for decision-making in pandemic preparedness. The work done in 2024 laid a strong foundation for co-creating strategies and tools to manage spillover risks more effectively in the future.

Work Package 5 (WP5)

WP5, led by Universitätsklinikum Heidelberg (UKHD), focused on developing a community-based Pandemic Prevention and Preparedness Literacy (3PL) intervention to strengthen pandemic readiness in Thai communities. In 2023, WP5 collaborated with WP1 to develop a comprehensive baseline survey tool to assess knowledge, attitudes, beliefs, and practices (KABP) related to spillover risks in Thai communities. This survey which incorporated interactive spatial mapping, was reviewed and validated by a panel of experts and approved by Chulalongkorn University's Ethics IRB. Additionally, WP5 prepared to implement the survey online and in the field while integrating findings from qualitative baseline data collection from WP1 to inform the 3PL intervention design.

In 2024, WP5 engaged in various PANDASIA activities, including fieldwork in collaboration with other WPs. WP5 also worked closely with WP1 and WP9 to ensure alignment of their activities across the project. In May 2024, a co-production workshop with NOVA Universitywhich was held in Lisbon from 7 to 8 May 2024, served as an opportunity for UKHD (represented by Kate Bärnighausen), QMUL, and NOVA researchers to work together, identify areas of collaboration, and co-develop a work plan. Later in December 2024, a hybrid co-production workshop took place in Heidelberg, Germany, bringing together researchers from WP1, WP5, and WP9. This workshop focused on enhancing collaboration among the work packages and aligning their activities and fieldwork plans for 2025.

From November 27 to December 8, 2024, WP5 (represented by Cássia Rocha from UKHD) participated in the fieldwork activities for social sciences in Chanthaburi. The WP5 team, composed of researchers from KKU, NMBU, UKHD and 11 research assistants from KKU, visited 35 villages in Pong Nam Ron and Soi Dao districts to present the project and conduct the quantitative data collection with around 680 participants. The comprehensive questionnaire used in this activity, developed by WP5 in collaboration with WP1, was aimed to collect key information for the 3PL intervention, such as pandemic prevention and preparedness knowledge, literacy, attitudes, beliefs and practices (KABP) among villagers. WP5's role in the fieldwork activities included training the team to use the Lime Survey online platform and ensuring on-site quality control of the data collected (Figure 10).

***Figure 10**. Training for the baseline quantitative survey data collection at Khon Kaen University.*

Work Package 6 (WP6)

The WP6, led by SUPA71 and with the support of the Leibniz Institute for Zoo and Wildlife Research (IZW), serves the communication, exploitation, and dissemination (CED) function of the project the needs of identified target audience and potential users of project results, thus ensuring project impacts of preventing the spread of emerging infectious diseases. In 2023, WP6 developed and submitted the Knowledge Exchange Strategy Report (D6.1) and Project Website and Social Media Report (D.6.1) in June 2023. The PANDASIA website and social media platforms, including Facebook, LinkedIn, and X (formerly Twitter), were launched to share updates, project milestones, and results with diverse audiences. WP6 also completed the first-year Communication Activity Report (D6.3) in December 2023, summarizing the progress of dissemination and engagement efforts. The WP6 team produced and shared communication materials, such as project flyers and a two-page synopsis, to make the project outcomes accessible to the scientific community, policymakers, industry stakeholders, and the public.

In 2024, WP6 held regular online team meetings to review communication strategies and the progress of planned activities. These meetings facilitated the production of videos, newsletters, press releases, and social media updates, as well as the dissemination of relevant information and messages on PANDASIA's website and social media platforms. The WP6 team also focused on updating and refining the knowledge-sharing/communication strategy, creating more infographics, and planning for a project glossary of technical terms and policy briefs.

During the year, WP6 produced and uploaded videos that highlighted the Stakeholders' Mapping Workshops. The first video highlighted the workshop with government stakeholders (October 17–18, 2023), while the second focused on civil society organizations (October 19, 2023). These videos aimed to enhance the local partners' awareness of PANDASIA's purpose and activities thereby getting their support of the project.

On February 26, 2024, WP6 released a video showcasing field data collection activity in various locations across Thailand. WP6 shared an interview with Dr. Overgaard, PANDASIA's Principal Investigator, conducted at Norway Life Science 2024, highlighting the project's goals and insights (Figure 11). In March 2024, WP6 disseminated two press releases. The first release announced PANDASIA's progress in understanding and preventing zoonotic spillovers and pandemics. The second press release emphasized the support of Thai communities in field data collection efforts. These releases were shared across consortium partners' press offices, social media platforms, in addition to the project website.

Figure 11. *Dr. Hans J. Overgaard, PANDASIA's Principal Investigator, during an interview at Norway Life Science 2024*

In April 2024, WP6 published newsletter volume no.3, detailing PANDASIA's field sample collections in Chanthaburi Province, Thailand. This newsletter provided updates to stakeholders on the project's ongoing activities and achievements. On May 31, 2024, WP6 uploaded a YouTube video titled PANDASIA Investigates Zoonoses with Wildlife and Non-Wildlife Origins in Thailand. The video highlighted the collection of wildlife, non-wildlife, and environmental samples, showcasing the multidisciplinary approach of the project. Between February to September 2024, WP6 focused on video production and dissemination to enhance the visibility of PANDASIA's work. The WP6 team finalized the PANDASIA image video clip, showcasing the project's key achievements and activities. This video was shared widely on various social media platforms to engage stakeholders and the broader public. Additionally, SUPA71 created and published videos on YouTube (Figure 12).

Figure 12*. PANDASIA's experts in testimony of the project's goals and activities*

In September 2024, WP6 consolidated a media contact list that spans the European, Thai and other Southeast Asian countries to facilitate effective communication and dissemination of PANDASIA's results.

From October to December 2024, WP6 planned important communication activities to improve the project's outreach engagement. WP6 team started preparing the PANDASIA glossary of technical terms, which will explain in local and simple language key project terminologies that can be understood by a non-technical individual. The glossary will be uploaded on the website and social media, helping audiences understand the project better. Towards the end of the year, WP6 will work closely with WP5 in analyzing the results of their survey from which we will draw recommended social and behavior change communication (SBCC) messages and materials for pre-testing and validation by the concerned local communities.

Work Package 7 (WP7)

Work Package 7 (WP7), led by the Norwegian University of Life Sciences (NMBU), focused on effective coordination and management of the PANDASIA project. In 2023, WP7 managed scientific coordination, internal communication, and project progress tracking, ensuring smooth implementation and adherence to agreed plans and ethical standards. Regular biweekly meetings were held with consortium partners to share updates and align activities across work packages. Key achievements included the development and delivery of the Data Management Plan (D7.1) for the sixth month of the project (M6). Additionally, the External Scientific Advisory Board (SAB) was established (D7.2), bringing together experts to provide guidance and oversight for the project. WP7 also facilitated financial and technical reporting to ensure compliance with grant agreements and managed legal, ethical, and contractual aspects of the project.

In 2024, WP7, led by NMBU, played an important role in managing and coordinating the various activities of the PANDASIA project. WP7 continually coordinated the work packages, organized scientific activities, communicated within the team, tracked project progress, managed finances and reports, ensured legal and ethical compliance, and handled data management. Throughout the year, coordination activities included both detailed discussions with specific work packages and broader meetings such as Steering Committee Meetings (SCMs) and the General Assembly. Two SCMs were held in 2024, with representatives from all partner institutions. These meetings followed up on the agreed-upon decisions of the consortium annual meeting held virtually on May 28–29, 2024. These meetings also allowed the WP7 team to review the first year's progress and plan for the future while saving time and resources, as many members were busy with fieldwork.

To support internal communication, WP7 organized biweekly meetings from the start of the project until March 2024. From April onward, monthly meetings were held, where each work package shared updates on their activities. NMBU also ensured all team members had access to a shared folder for important documents. In terms of financial and technical reporting, WP7 prepared and submitted the first periodic report covering January 2023 to June 2024. NMBU guided the partners on financial rules, explaining how to calculate costs and use the EU's reporting portal (SYGMA). They also held group and one-on-one meetings to help partners correctly report their work.

In 2024, WP7 ensured that project activities followed legal and ethical standards. NMBU ensured compliance with the European Commission's guidelines, managed amendments to the project plan, and addressed any ethical or legal issues. In May 2024, NOVA joined the consortium, adding to the project's pool of expertise. WP7 also ensured that all research activities received the necessary ethical approvals before starting the research on humans, domestic animals, and wild animals.

For data management, WP7 worked on implementing the Data Management Plan (DMP), which was created in 2023 with help from WP5. This plan provided clear rules for collecting, storing, and sharing data while following European data protection laws (GDPR). The PANDASIA project has generated a variety of outputs, such as data, software, and research articles. WP7 ensured these outputs followed FAIR principles, meaning they are easy to find, use, and share. Many project models became open access in 2024, with plans to make even more available, so the project's work could benefit a larger audience.

Work Package 8 (WP8)

According to the first 18-month report of PANDASIA, Work Package 8 (WP8) focused on ensuring compliance with global biosafety and biosecurity standards, managing ethics and administrative approvals, and addressing ethical considerations raised by the project. In 2023, the Ethics Review Board (ERB) was established, consisting of Prof. Dr. Singhasivanon, the

Secretary General of the SEAMEO TROPMED Network. The composition of the ERB, along with its mandate and expected contributions to PANDASIA, was submitted to the EU in June 2023 as part of Deliverable D7.6. In 2024, the engagement of Prof. Dr. Singhasivanon continued as the sole Ethics Advisor for the consortium. Further information on the coordination and communication process with the Ethics Advisor is detailed in Deliverable D8.1. WP8 has remained committed to maintaining ethical standards and transparency in all project activities.

Work Package 9 (WP9)

According to the first 18-month report of PANDASIA, Work Package 9 (WP9) led by NOVA, serves in One Health interventions and community engagement and officially joined the PANDASIA project in 2024. NOVA, being new member of the consortium, prepared and obtained ethical approval from the Ethics Committee of NOVA IMS and the MagIC Research Centre in May 2024. The next step is to obtain approval from regulatory authorities in Thailand. To support this, NOVA is finalizing a detailed document that explains the project's goals, methods, potential risks, and other required materials for ethical review in social and behavioral science research.

One of NOVA's main tasks in 2024 was to create a relationship diagram to map the factors involved in potential zoonotic spillover risks. This diagram would show how environmental and social factors, community behaviors, and policy structures are connected, enabling to identify key areas for intervention. This work is ongoing and will guide future project activities.

As of this reporting period, NOVA is also developing a protocol to assess Village Health Volunteer (VHV) training. This protocol will measure how the training improves knowledge, skills, and attitudes, as well as how well volunteers remember the information over time. Surveys are being designed to collect this data.

In addition, NOVA is planning a community empowerment initiative in the PANDASIA study sites. This will focus on improving knowledge and involvement in One Health interventions. As NOVA prepares to study citizen behavior to understand what influences people to take part in pandemic prevention activities and adopt protective habits, it currently is refining the survey tools and working with partners to finalize them. The ethical approval process for data collection is ongoing. After obtaining the approval for data collection, NOVA can start these activities.

3.2 Communication Activities of Work Package 6

According to WP6's activities under section 3.1 Relevant Activities from PANDASIA Consortium Partners, communication activities of WP6 in 2024 6 in 2024 focused on continuing and improving the work started in 2023 to enhance the dissemination, exploitation, and communication of PANDASIA's activities and the results were as follows:

Knowledge Exchange Strategy Report Updated: In 2024, planned activites reviewed and revised the 2023 communication strategy. In particular, WP6 reflected on achieving the success indiactors that were promised in 2023.

PANDASIA website and social media platforms maintained: The PANDASIA website which was launched (https://pandasia-project.com/) in 2023 was maintained. SUPA71 continuously uploaded materials on social media platforms such as Facebook, LinkedIn, Twitter, and YouTube. Currently, the website and social media platforms have been maintained and updated following PANDASIA's communication guidelines. The engagement statistics from May to December 25, 2023, were as follows:

- **Website engagement**: As of December 28, 2024, the website had a total of 21,615 page views. The website received visits from various countries, including Thailand, Cambodia, United States, Lao PDR, Germany, Vietnam, Portugal, Norway, Philippines, United Kingdom, Sweden, Indonesia, Bangladesh, Costa Rica, China, Canada, Russian Federation, Serbia, Netherlands, Spain, Belgium, Australia, Hungary, Singapore, Brazil, Denmark, Poland, Japan, India, Italy, Hong Kong, Turkey, Austria, Isle of Man, Israel, France, Colombia, Republic of Korea, Nepal, Greece, Ghana, Mexico, Finland, Zambia, Malaysia, Ireland, Czech Republic, Zimbabwe, South Africa, Venezuela, Taiwan, Slovakia, Saudi Arabia, Papua New Guinea, Peru, Panama, Nigeria, Republic of Moldova, Saint Kitts and Nevis, Kenya, Islamic Republic of Iran, Croatia, Guatemala, Gambia, Ethiopia, Cape Verde, Switzerland, and Argentina (Figure 12).

Figure 13. *Map of PANDASIA's website engagement*

- **Outbreak News:** As of December 28, 2024, a total of 43 outbreak news updates (Figure 14) were published on the PANDASIA website.

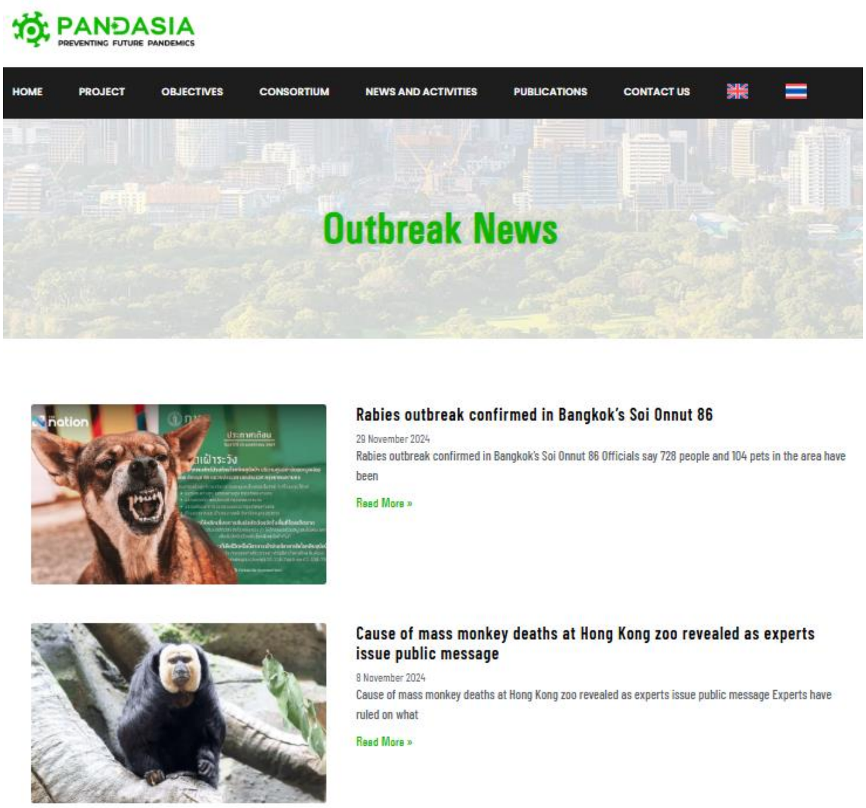

Figure 14*. Photos of the sample outbreak news*

- **Activities News:** As of December 28, 2024, a total of 12 activities news (Figure 15) were published on the PANDASIA website and social media platforms, including Facebook, X (formerly Twitter), and LinkedIn.

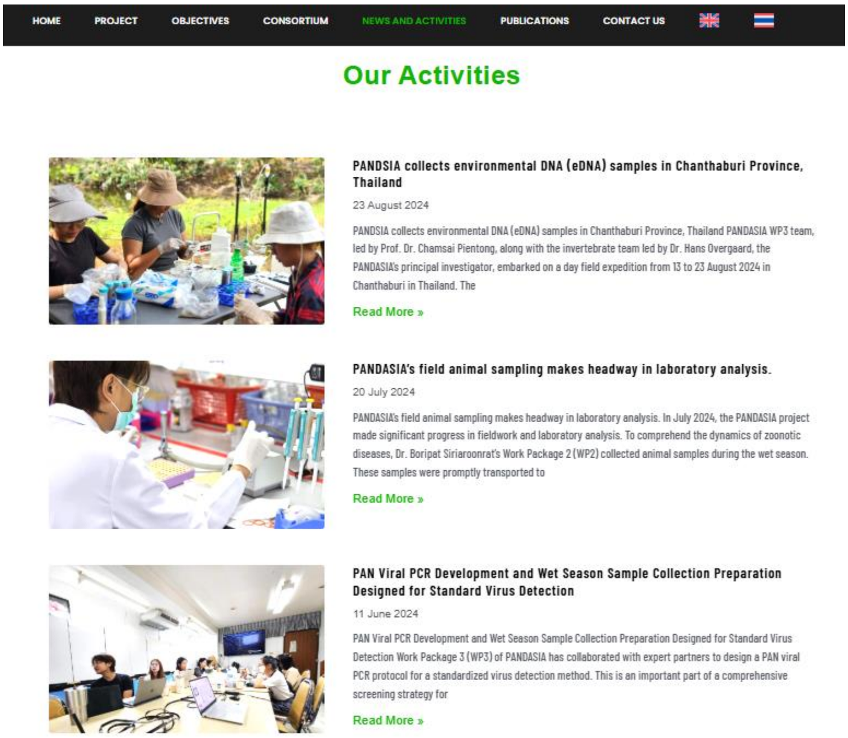

Figure 15*. Photos of activity news*

- **Event Posts:** As of December 28, 2024, a total of 30 event posts (Figure 16) were shared across PANDASIA's social media platforms, including Facebook, X (formerly Twitter), and LinkedIn, engaging a wider audience.

Figure 16. Photos of events posts

- **Facebook:** As of December 28, 2024, there were 6,400 page likes and followers, and 52 posts shared of the activities, and around 40,000 reached the online Facebook audiences (Figure 17).

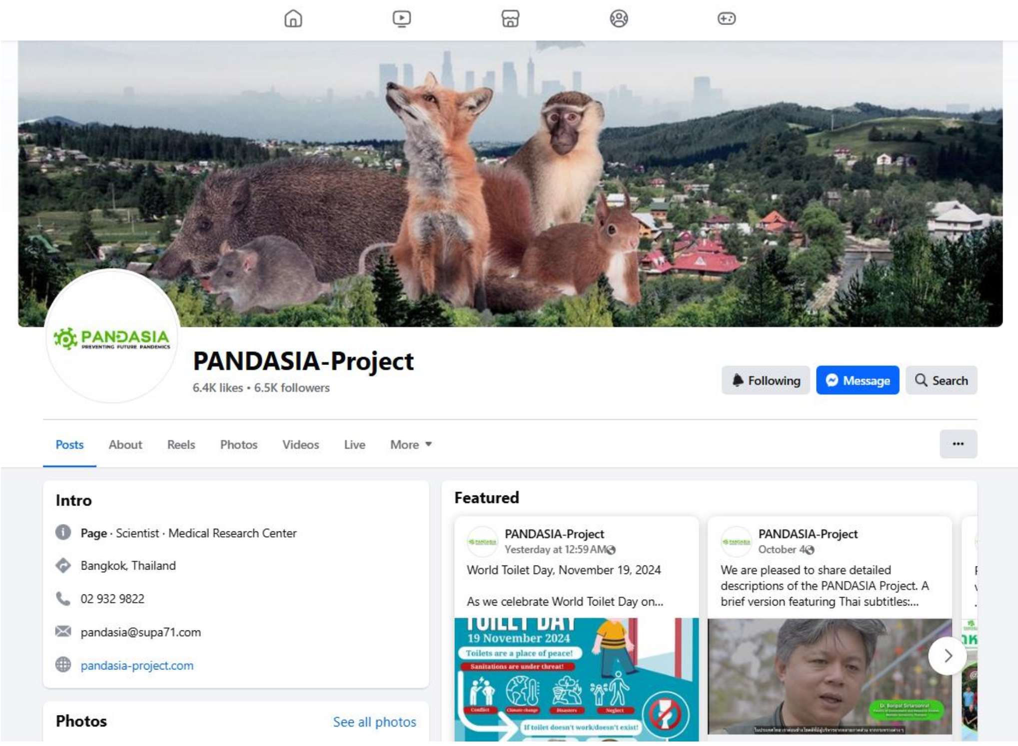

Figure 17. PANDASIA's Facebook page

- **LinkedIn**: As of December 28, 2024, there were 1,402 connections, 1,394 followers, 235 posts and reposts (Figure 18).

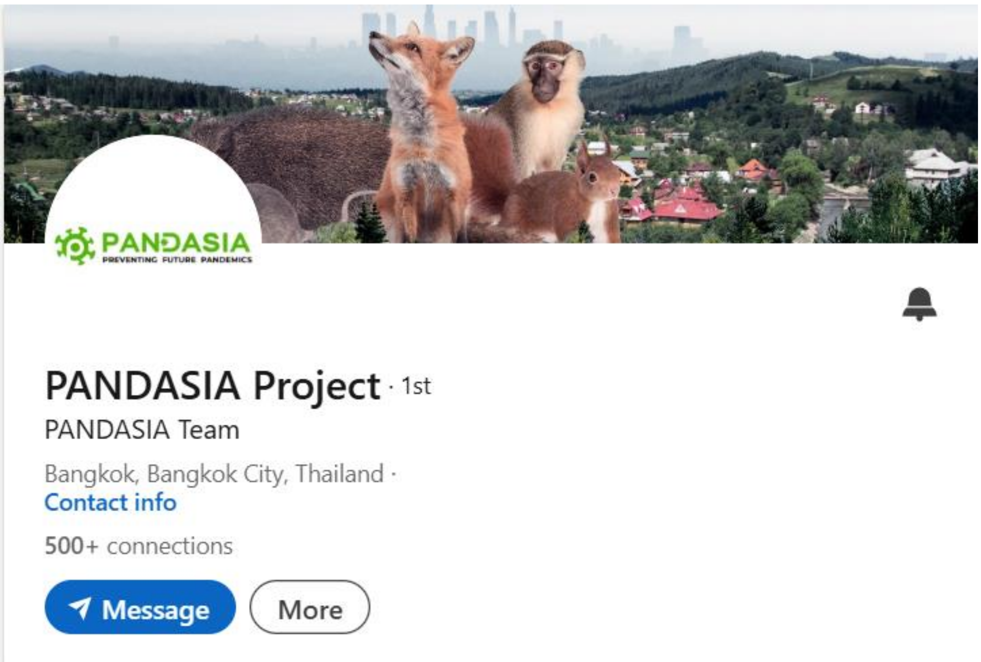

Figure 18. *PANDASIA's LinkedIn*

- **X (formerly Twitter)**: There were 450 posts and re-posts from June 2023 to December 2024. We followed 739 professionals and projects and received 97 active followers from various professional fields, especially in public health, environment, and communication (Figure 19).

Figure 19. *PANDASIA's X*

Communication materials: As of December 28, 2024, communication materials were downloaded 495 times in various documents. The most downloaded communication materials included Press Releases (PRs), Newsletters, Project Synopses, and Project Flyers. The communication materials prepared and produced by the WP6 team in English and Thai included:

- **Press release**: Three press releases were published and distributed to relevant stakeholders.
- **Flyers**: PANDASIA flyers in English and Thai that introduced the project descriptions that were posted on the website.
- **Newsletters:** Three newsletters that were produced and distributed to the public and relevant stakeholders. Apart from serving the Consortium members' information needs, the periodic newsletter narrated the progress of project activities to our external audience when uploaded on the website.
- **Internal Communication Guideline shared:** The WP6 team initiated a guideline on internal communication and shared this among the Consortium members, providing guidance when taking photos of and getting anecdotes from local people. The guidelines would ensure confidentiality of information from local people and uphold values like trust, openness, positive thinking, accountability, and equality. The document also detailed the procedures for file sharing, media relations, and documentation of meetings and events. It was designed to foster harmonious working relationships and ensure the success of project goals through clear and coordinated communication among diverse consortium members.

Communication with press and media contacts in Europe and Thailand initiated: The WP6 team started to connect with the press offices of consortium members. For any results or activities that interest the members and the external audience, the WP6 team would gather and share such information via email, any other channel, or an online platform.

3.3 Technical Conferences and Meetings with Relevant Stakeholders

Technical conferences and meetings are ideal avenues to share the work of PANDASIA, meet other experts/scientist working on the same field or issues, and learn from others' work. In 2024, consortium members actively joined local and international conferences as follows:

Work Package 1 (WP1)

In 2024, WP1 actively participated in key PANDASIA events and collaborative efforts to maintain alignment and ensure project progress. WP1 joined a two-day hybrid workshop held at Umeå University on February 26–27, 2024, which focused on advancing WP4 modeling efforts and strengthening collaboration among WP1-4. Additionally, WP1 regularly attended PANDASIA's monthly meetings, which served as a platform for sharing updates on the progress and activities of each work package. Moreover, WP1 attended the annual PANDASIA meeting, held virtually on May 28–29, 2024. The meeting brought together all project partners to review the progress made during the first year and plan for the next phase. This event provided an opportunity for

work packages to align their activities and ensure coordinated efforts across the consortium for better collaboration and setting a clear direction for future tasks.
The WP1 experts from QMUL and UKHD joined two significant co-production workshops facilitated collaboration and alignment within the PANDASIA project. The first workshop, held in Lisbon on May 7–8, introduced NOVA as the newest project partner and focused on identifying collaborative opportunities and developing a joint work plan for WP1 and WP9, ensuring NOVA's seamless integration into the consortium. The second workshop, hosted in Heidelberg from December 9–13 and led by QMUL and UKHD, presented the results of stakeholders' mapping data analysis to WP5. These findings were instrumental in informing the development of targeted interventions and translating research outputs into actionable strategies, thereby ensuring alignment with the overarching project objectives.

Regular internal meetings were held throughout the year among WP1 team members, including CU, QMUL, UKHD, and the PI from NMBU. These meetings were instrumental in reviewing updated activities and planning the transition of CU's tasks to QMUL and UKHD after June 2024. This handover ensured continuity in WP1's responsibilities and smooth execution of the project.

Work Package 2 (WP2)

In 2024, WP2, led by KKU, actively participated in key PANDASIA events, including the WP4 workshop held on February 26–27 at Umeå University. This two-day hybrid workshop focused on advancing WP4 modeling efforts and strengthening collaboration across WP1-4. Discussions included identifying relevant processes and parameters for models, aligning data collection with modeling needs, and fostering interdisciplinary feedback. WP2 also attended PANDASIA's monthly meetings and the annual meeting on May 28–29, which provided platforms for updates and coordination across work packages. Additionally, WP2 held internal team meetings and collaborated closely with NMBU and KKU to plan and organize field activities, ensuring smooth execution of tasks and alignment with project objectives.

Work Package 3 (WP3)

WP3 regularly met with key partners in support to achieving its goals, encourage teamwork, and ensure smooth project progress. These meetings were essential for making technical improvements, keeping communication clear, and hitting project targets on time.

WP3 worked closely with Talenome DNA Professional Co., Ltd. (TALENOME), a trusted partner known for their expertise in molecular diagnostics for healthcare. The main goal of these meetings was to develop and improve a multiplex PAN-viral qRT-PCR system, an important tool for detecting multiple viral families in a single test. The collaboration included regular virtual meetings and in-person sessions, ensuring steady progress and practical testing of the system.

Online meetings were held to share updates, solve technical problems, and align strategies. TALENOME also sent representatives to Khon Kaen University (KKU) to help with hands-on lab testing. This teamwork helped combine advanced technology with practical use, ensuring that the system meets both scientific and operational needs.

WP3 also collaborated regularly with WP2, the invertebrate team. These meetings were held consistently throughout the year, averaging at least one meeting per month, ensuring effective communication and coordination across all activities. These meetings focused on planning fieldwork and analyzing samples. They were key to organizing sample collection during both dry and wet seasons and ensuring all targeted areas were covered.

The WP3 team shared findings from past fieldwork, exchanged sample data, and discussed better ways to process and analyze samples. Prof. Dr. Greenwood provided valuable global insights, especially on virus enrichment and sequencing techniques. These meetings also helped solve logistical issues, improve field strategies, and make sure fieldwork and lab analyses worked well together.

Work Package 4 (WP4)

In 2024, WP4 carried out a variety of important activities to support the PANDASIA project, focusing on collaboration, research dissemination, and policy engagement. From 26 February to 29 February 2024, the WP4 team organized a WP4 workshop in Umeå, Sweden. This workshop brought together participants for presentations and collaborative discussions, setting the stage for productive partnerships within the project. Later, on 3 April and 4 April 2024, WP4 attended the ERC Policy Meeting in Oslo, Norway, contributing to policy-related discussions that align with PANDASIA's goals.

From 27 October to 29 October 2024, WP4 participated in the Multicrisis Conference at the Norwegian Pandemic Centre in Bergen, Norway. The WP4 team provided a presentation and engaged in discussions on critical topics, sharing insights and exploring opportunities for collaboration. A picture from this presentation was also shared to highlight the team's active involvement.

WP4 contributed to the academic community through the publication of a research paper in February 2024, titled *"Season of death, pathogen persistence, and wildlife behavior alter number of anthrax secondary infections from environmental reservoirs"*. Published in the *Proceedings of the Royal Society of London. Biological Sciences*, this paper showcased WP4's contributions to advancing scientific understanding.

The WP4 team maintained regular communication through monthly and bimonthly meetings, as well as ad hoc and steering group sessions. WP4 also collaborated closely with WP2 and WP6 during these meetings to ensure alignment across the project. On 15 November 2024, the WP4 team presented a preliminary chapter contribution to the upcoming Routledge book *"From Farm to Pandemic"* during an online seminar.

Work Package 5 (WP5)

In February 2024, WP5 researchers attended the workshop in Umeå University, Sweden, organized by WP4 on February 26–27, 2024 (Figure 20). This workshop opened the discussion with other WPs on the relevant processes and parameters integrating the model developed by WP4 team.

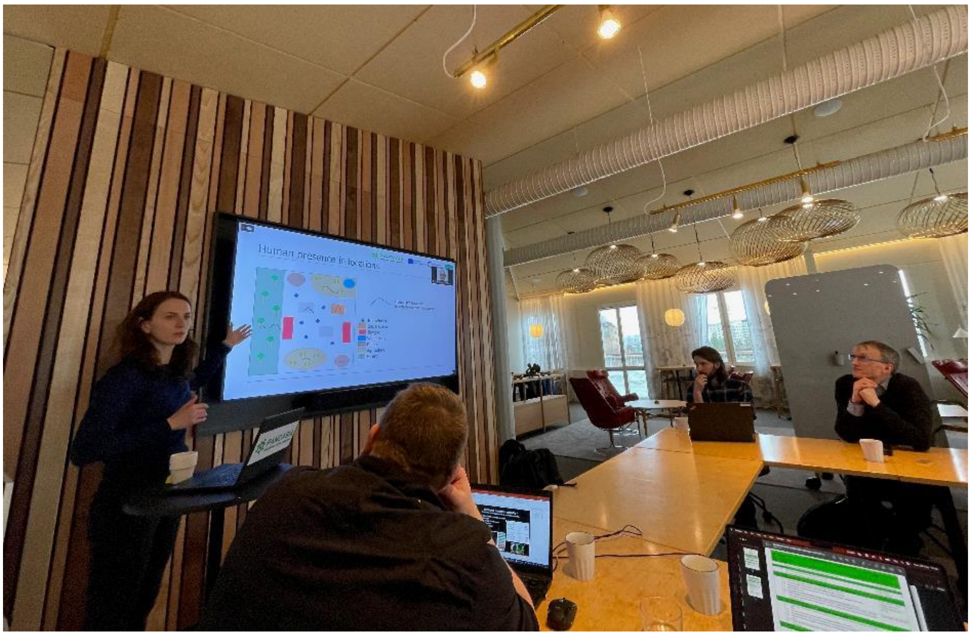

Figure 20. *Marina Treskova (UKHD) presenting at the WP4 workshop in Umeå University, Sweden, February 26–27, 2024*

Additionally, WP5 participated in the PANDASIA annual meeting held virtually on May 28–29, 2024, which reviewed the project's progress and aligned future plans. WP5 also regularly attended PANDASIA's monthly meetings, which served as platforms for sharing updates and ensuring coordination across the consortium.

Work Package 6 (WP6)

WP6, led by SUPA71 participated in several conferences and meetings both online and onsite, as follows:

WP6 participated in WP4 workshop on February 26–27, 2024 and joined the PANDASIA annual meeting that was held virtually on May 28–29, 2024. WP6 also regularly attended PANDASIA's monthly meetings for sharing updates and ensuring coordination across the PANDASIA consortium partners.

On October 8, 2024, the SUPA71 team participated in the 24th International Conference of Public Health Sciences at Chulalongkorn University in Bangkok, Thailand. Held at the Mandarin Ballroom, Mandarin Hotel, the conference was themed "Public Health 360: From Policy to Practice" and covered a wide range of topics, including global health, disease prevention, aging, mental health, sexual and reproductive health, healthcare management, universal health coverage, environmental and occupational health, digital health technologies, and innovative health sciences. Experts gathered to discuss and share knowledge aimed at improving public health systems and addressing key global health challenges (Figure 21).

Figure 21. *The 24th International Conference of Public Health Sciences at Chulalongkorn University in Bangkok, Thailand, October 8, 2024*

From November 20–22, 2024, SUPA71 participated online in the ESCAIDE 2024 conference, a hybrid event hosted in Stockholm and virtually. This prominent conference brought together experts and stakeholders in applied infectious disease epidemiology to discuss emerging health threats, surveillance, and prevention strategies. The program included plenary sessions such as "Prevention vs. Cure: What Can We Learn from Cancer, Crime, and Climate Change?" and "Disease X: Are We Ready?", fireside sessions on topics like vaccine-preventable diseases and antimicrobial resistance, and numerous poster tours covering surveillance, outbreaks, and public health strategies. The event also featured experts' panel discussions, networking opportunities, and side sessions on public health innovations and methodologies (Figure 22).

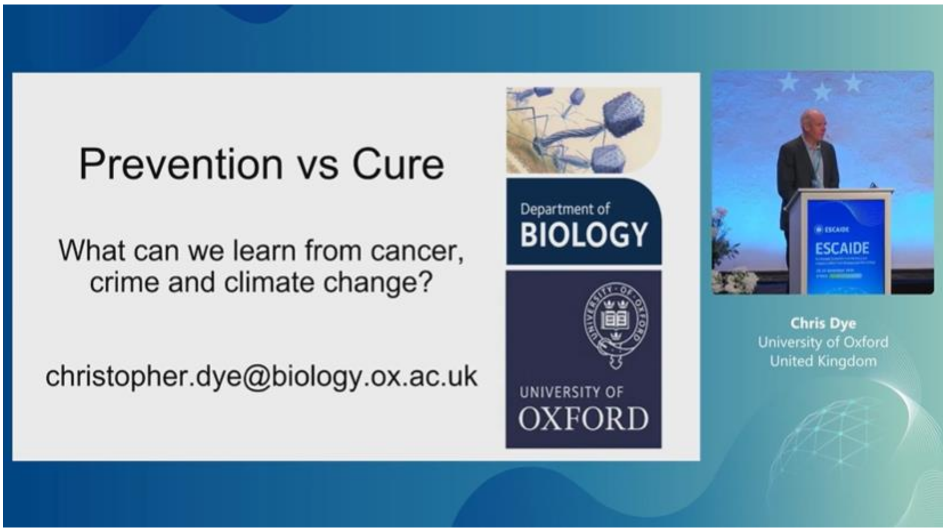

Figure 22. *The ESCAIDE 2024 Conference, November 20–22, 2024*

On November 26, 2024, SUPA71 participated in the EU Health Policy Platform (EUHPP) Annual Meeting 2024. This meeting provided an opportunity for health stakeholders to discuss and endorse the 2024 Joint Statements developed by the four Thematic Networks. Over the past months, these networks collaborated through webinars and discussions to draft their position papers, which were presented during the meeting with the discussions, supporting the review and potential endorsement of the Joint Statements, aligning with the project's commitment to health policy engagement and collaboration (Figure 23).

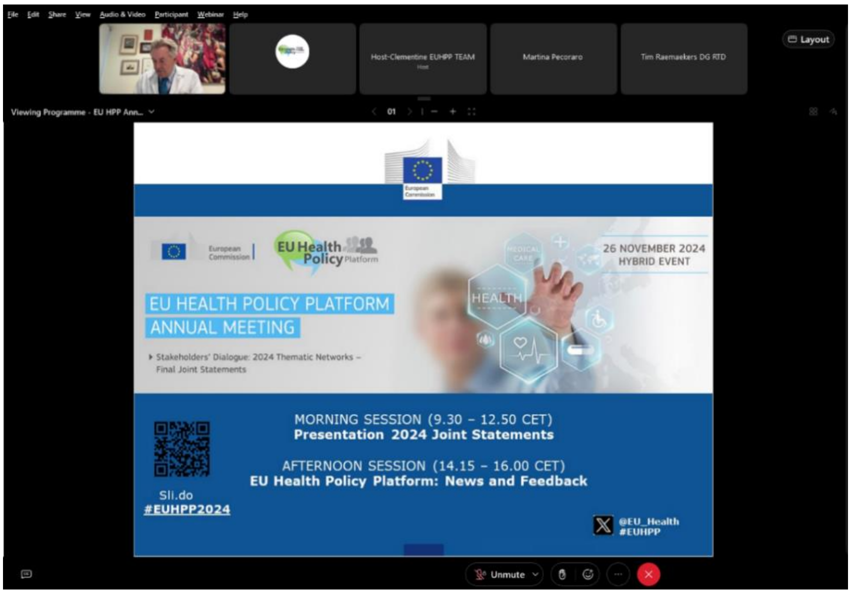

***Figure 23**. EU Health Policy Platform (EUHPP) Annual Meeting 2024, November 26, 2024*

On November 28, 2024, WP6 participated online in the WHO EPI-WIN Webinar titled One Health Intelligence: Global Collaboration for Emerging Health Threat Detection, the webinar focused on the Global Early Warning and Prevention System (GLEWS+), which identifies and monitors disease events to mitigate risks related to emerging, endemic, and epidemic diseases, as well as environmental health and food security threats (Figure 24).

***Figure 24**. WHO EPI-WIN Webinar on One Health Intelligence: Global Collaboration for Emerging Health Threat Detection, November 28, 2024*

Work Package 7 (WP7)

In 2024, WP7 facilitated the coordination and collaboration of the PANDASIA project through regular monthly meetings and the organization of the annual meeting. The monthly meetings provided a platform for each work package to report progress, discuss challenges, and align activities across the consortium. The annual meeting, held virtually on May 28–29, 2024, brought together all partners to review the first year of the project and plan for the upcoming phases. These meetings ensured smooth communication and efficient decision-making within the consortium.

As the coordinator of PANDASIA, NMBU actively communicated the project's objectives and engaged in scientific discussions on zoonotic spillover and pandemic preparedness. Dr. Overgaard, the Principal Investigator of PANDASIA, participated in multiple important events throughout 2024. He visited Heidelberg in February and October for general project planning meetings, focusing on aligning activities across work packages and refining the overall project strategy. Dr. Overgaard presented the project at the Norway Life Science 2024 conference in Oslo in February 2024 (200 participants). These presentations emphasized the importance of upstream surveillance strategies and highlighted PANDASIA's contributions to pandemic prevention and preparedness. Additionally, Dr. Overgaard traveled to Umeå University for the WP4 workshop on February 26–27, 2024, where discussions focused on data collection methodologies, analysis frameworks, and strengthening inter-WP collaboration (Figure 25). These visits enhanced coordination and facilitated more integrated planning among the partners.

Figure 25. *Dr. Hans J. Overgaard Participating in the WP4 Workshop at Umeå University, February 26–27, 2024*

WP7 also emphasized participation in conferences and workshops to share insights, foster collaboration, and showcase the project's progress. Dr. Overgaard, along with Prof. Dr. Rockloev and Dr. Treskova (WP5, UKHD), Prof. Dr. Greenwood (IZW), Dr. Phantanawiboon

(WP3, KKU), and Dr. Jairak (WP2, MU), attended the Global Issues Convention 2024, hosted by the Volkswagen Foundation (https://www.volkswagenstiftung.de/en/events/global-issues-convention-2024). Furthermore, WP7 also participated in the conference of the Global Issues Convention 2024, which had experts from various disciplines, provided an opportunity to highlight PANDASIA's contributions to addressing global challenges in pandemic preparedness and zoonotic spillover risk mitigation. The participation of multiple WP representatives underscored the project's transdisciplinary approach and its relevance to global issues.

Work Package 8 (WP8)

WP8, led by Prof. Dr. Singhasivanon, the Secretary General of the SEAMEO TROPMED Network, provided valuable guidance on research ethics and collaborative practices within the PANDASIA project. The WP8 joined a meeting with KKU, MU, CU, and NMBU to discuss ethical considerations and support effective teamwork across work packages. In the meeting, the WP8 emphasized the importance of adhering to ethical standards in the project activities and provided recommendations to ensure smooth collaboration and alignment with the project's objectives.

Work Package 9 (WP9)

Work Package 9 (WP9) focused on collaboration and aligning tasks with other work packages in the PANDASIA project. On 7 May 2024 and 8 May 2024, a meeting was held in Lisbon with WP1 and WP5, attended by seven participants. This in-person meeting was important for discussing progress, identifying gaps, and aligning the work between these teams. On 28 May 2024 and 29 May 2024, WP9 also attended the annual PANDASIA online meeting.
On 4 June 2024 and 24 June 2024, WP9 met with WP1, WP4, and WP5 to discuss specific tasks (T9.1, T9.4, and T9.5) and ensure everyone was working toward the same goals. On 10 June 2024, the Steering Committee met to provide updates on governance and project planning, and on 14 June 2024, WP9 discussed ethical procedures and data collection with the project coordinator.

Monthly project meetings were held on 8 July 2024, 9 September 2024, 23 October 2024, and 11 November 2024, bringing all partners together to share updates and review progress. On 10 September 2024, WP9 also discussed subcontracting procedures and timelines for data collection and met with WP5 to align work activities. Additionally, on 24 October 2024, WP9 and WP5 discussed task T9.3 in detail, with six participants attending. These activities kept WP9 on track and ensured smooth coordination across the project

4. Impact

As presented in the first 18-month report of PANDASIA, the PANDASIA partners will publish in high-impact, open-access journals and participate in regional and global conferences on topics such as emerging infectious diseases, pandemic preparedness, and One Health. Scientific impact will be disseminated and shared through publications, presentations at international meetings, and collaboration with leading scientists. Among consortium partners, knowledge sharing through publications and conference presentations, has been discussed to exchange ideas and advance research methodologies.

WP1. Society, Livelihoods & Pandemic Risk Perceptions

Health messages and community engagement. The project supported Ms. Vedlog Kveen, an MSc student at NMBU, in completing her thesis, "Rapid Review of Public Health Messaging and Community Engagement Initiatives during the COVID-19 Pandemic in the Greater Mekong Subregion," in May 2024. Her review, based on 26 articles, highlighted Vietnam's creative risk communication methods, such as songs and slogans, and Thailand's village health volunteer network, which effectively managed early pandemic responses but faced challenges in reaching marginalized groups like migrants. The study emphasized the importance of designing inclusive communication programs with community input to ensure accessibility and effectiveness. The thesis will be published in Global Health Action and submitted as **D1.3 (Task 1.1).**

Zoonotic spillover and community stakeholder. The project supported Ms. Reien, an MSc student at NMBU, in completing her thesis, "The Risk of Zoonotic Spillover: A Qualitative Study of Community Stakeholders in Rural Thailand," in May 2024. Using data from fieldwork conducted in Chanthaburi in October-November 2023, her study included 26 in-depth interviews and shared walks with community stakeholders to explore locations and behaviors linked to zoonotic spillover risk. Key findings identified high-risk locations such as homes, villages, farms, schools, markets, and forests. Risk factors included consuming raw animal products, living near animals, transferring animal products across borders, and environmental changes. The study highlighted the importance of developing community-informed prevention strategies tailored to the realities of human-animal-environment interactions. Her thesis will be submitted to Global Health Action **(Task 1.1).**

Wildlife interactions. WP1 prepared a manuscript in collaboration with Chiang Rai Provincial Health Office about wildlife interaction characteristics among villagers in the two study sites in Chiang Rai province. The results will be useful for understanding villagers' risk perceptions regarding previous/future pandemics and spillover. In addition, the findings will be the basis for developing social and behavior change messages and materials for community interventions in identified areas in Thailand.

Zoonotic spillover and community stakeholders. The project supported Ms. Reien, an MSc student at NMBU, in completing her thesis, "The Risk of Zoonotic Spillover: A Qualitative Study of Community Stakeholders in Rural Thailand," in May 2024. Using data from fieldwork conducted in Chanthaburi during October-November 2023, her thesis will be submitted to the journal Global Health Action **(Task 1.1).**

WP2. Wildlife, Livestock & Landcover Change

Mosquito data. In April 2024, WP2 started summarizing results from invertebrate collections for a peer-reviewed article titled "Mosquitoes Collected in Different Zoonotic Interfaces in Northern and Eastern Thailand" **(Task 2.4).**

Landscape structure. WP2 started developing a systematic map protocol to examine the global evidence on how landscape structure influences zoonotic spillover risk. The objectives were to identify landscape characteristics driving spillover risk, and assess how changes in landscape structure affect the prevalence and spread of zoonotic diseases. The findings will support a future paper exploring the relationships between land cover, climate, and undifferentiated fevers over the past 20 years in northern and eastern Thailand **(Task 2.6).**

WP3. Virology

WP3 reviewed zoonotic viruses in domestic animals and studied the diversity and distribution of these viruses in bats and rats across Southeast Asia. This work included analyzing existing research and sequence data from a curated virus database to understand the prevalence and geographic spread of these viruses in key reservoir hosts. The findings will be published in two peer-reviewed articles: "Diversity of Zoonotic Viruses in Domestic Animals in Southeast Asia and Cross-Species Transmission Potential between Canines and Humans" and "Diversity and Distribution of Zoonotic Viruses in Bats and Rats in Southeast Asia." These studies provide insights into zoonotic virus transmission and inform public health strategies.

WP4. Mathematical Modelling of Spillover Risks

Virulence. WP4 prepared a manuscript on the potential drivers of virulence in respiratory viruses, such as COVID-19 or HPAI. The work included a conceptual model with promising preliminary results, and manuscript preparation is underway. This research aimed to improve understanding of the evolutionary factors influencing pathogen virulence, a critical aspect of pandemic potential.

Bat distribution. A manuscript was being prepared to model the distribution of fruit bats, a key reservoir host species, and their associated spillover risk. Understanding the spatio-temporal distribution of these hosts is critical for estimating spillover risk and the pathogens they may transmit to humans. Given the challenges in gathering detailed spatio-temporal data, accurate models will play a vital role in predicting and managing spillover risks.

Evolution. A manuscript on evolution modeling to explore the processes involved in pathogen transmission between host species was prepared. These evolutionary processes, which are not well understood, are likely critical for assessing spillover risk. The paper was aimed to identify key drivers and outcomes of these processes to enhance understanding of spillover dynamics.

Scientists of the PANDASIA consortium presented the following:

- *Virulence in the Anthropocene.* Authors: Kyrre Kausrud (NVI), Henrik Sjödin (UMU), Hans J. Overgaard (NMBU), Wendy C. Turner. Conference: *Long term consequences of the Covid-19 pandemic for society,* Pandemic Centre at the University of Bergen, Norway on 27. October 2023. This was an international conference with about 250 participants, held in English.
- *Within- and between-host processes of phenotype dynamics of zoonotic pathogens.* Authors: Henrik Sjödin (UMU), Kyrre Kausrud (NVI), Alex Greenwood (FVB-IZW) et al. Conference: Unifying the epidemiological and evolutionary dynamics of pathogens. Conference: *Unifying the epidemiological and evolutionary dynamics of pathogens*, The Nordic Institute for Theoretical Physics (NORDITA), Stockholm, June 2023. Previous modelling work related to PANDASIA.
- *Pathogen evolution and emergence in zoonotic spillover events.* Authors: Peter Fransson (UKHD), Henrik Sjödin (UMU), Kyrre Kausrud NVI. Conference: *Unifying the epidemiological and evolutionary dynamics of pathogens*, The Nordic Institute for Theoretical Physics (NORDITA), Stockholm, June 2023. Modelling work related to PANDASIA.
- *The evolution of infectious diseases: connecting within- and between-host processes.* Authors: Willian Silva (UMU), Åke Brännström (UMU), …, Henrik Sjödin (UMU). Conference: *Unifying the epidemiological and evolutionary dynamics of pathogens*, The Nordic Institute for Theoretical Physics (NORDITA), Stockholm, June 2023.

WP5. Pandemic Prevention and Preparedness Literacy (3PL) Intervention

One Health Content Review. As part of developing the educational intervention, WP5 reviewed and evaluated existing materials on zoonotic infection risks and pathogen spillover. The review covered resources from international organizations and agencies within Thailand. A manuscript summarizing the findings was drafted for peer review and future publication, highlighting gaps and needs in One Health communication materials.

Development and Expert Review of a Questionnaire for the Quantitative Survey. WP5 developed a questionnaire for the baseline quantitative survey in PANDASIA, drawing on qualitative studies and existing tools. The goal was to create methods for measuring zoonotic spillover risk dynamics by examining human-animal-environment interactions and societal factors. The questionnaire focused on assessing human-animal interfaces, spatio-temporal overlaps between people and reservoir host animals, and the frequency and types of contact. The objectives of the questionnaire are to:

- Measure human temporal presence at land-use human-animal interfaces resulting from routine activities.
- Quantify animal-human contacts at these interfaces.
- Gather baseline information on zoonotic spillover knowledge, behaviors, and self-efficacy before implementing the PANDASIA intervention.
- Identify the most commonly used communication channels.

WP5 engaged eight external experts, including scientists from Europe and Thailand, to review and evaluate the questionnaire. The evaluation assessed the questionnaire's necessity, clarity, relevance, and potential biases. Based on their feedback, WP5 revised the questionnaire and conducted a field test in Thailand. A manuscript is being planned to detail the questionnaire design, expert review process, and field test results

Human behaviour-induced zoonotic pathogen spillover in human populations and livestock worldwide: a scoping review. WP5 conducted a scoping review to identify household-level behaviors and practices that influence the risk of zoonotic spillover. The review protocol was registered on the Open Science Forum (osf.io) with the DOI: 10.17605/OSF.IO/3AN6T. This publicly accessible protocol outlined the objectives and methods, providing transparency and an overview of WP5's approach to this important topic.

WP6. Communication, Exploitation, and Dissemination

WP6 made significant progress in ensuring that PANDASIA's findings reach a wide range of audiences and stakeholders.

WP6 developed and updated a comprehensive knowledge exchange strategy aimed at improving the communication indicators, as well as improving internal and external communication across the PANDASIA consortium partners. This effort ensured the effective communication, exploitation, and dissemination of project findings to diverse audiences, including local communities, policymakers, and the scientific community.

WP6 established a strong project identity through the creation of branding materials, a project website, and active social media channels. These platforms reached a broad audience, including policymakers, civil society organizations, and the general public, helping to raise awareness about PANDASIA's goals and activities.

In 2025, WP6 will prioritize engaging with local communities in Thailand to ensure the sustainable use of project results. Planned collaboration with district authorities and village health and veterinary volunteers will make PANDASIA's findings accessible and actionable at the community level, strengthening pandemic preparedness and response strategies.

Furthermore, WP6 will support the dissemination of research findings through PANDASIA website, social media platforms, and publications in peer-reviewed journals. These efforts will ensure that PANDASIA's work contributes to global discussions on pandemic preparedness, One Health, and EcoHealth, enhancing its scientific impact.

WP7. Coordination and Management.

As the coordinator, NMBU is dedicated to communicating the PANDASIA project across various platforms to diverse audiences while engaging in scientific discussions on zoonotic spillover processes and their implications. In line with this commitment, the Principal Investigator, Hans J. Overgaard, presented the project and related topics at several scientific meetings and conferences:

- *The PANDASIA project - Understanding zoonotic spillover pathways for pandemic preparedness.* Centre for Pandemics and One Health Research (P1H) at the University of Oslo (UiO), Holmen Fjordhotell, 15 June 2023. Approximately 80 participants. Held in English.
- *The PANDASIA project - Understanding zoonotic spillover pathways for pandemic preparedness*. The Regional Community Forestry Training Centre for Asia and the Pacific (RECOFTC), Kasetsart University, 26 October 2023, Bangkok, Thailand. Approximately 15 people were present, held in English.
- *Preventing pandemics at the source - Need for upstream spillover surveillance strategies.* Norway Life Science 2024, Oslo, 13 Feb 2024. Innovations in Pandemic Preparedness: Bridging Epidemiology and Public Health for a Safer Future. Approximately 200 participants. Held in English.

5. Update of the plan for exploitation and dissemination of results (if applicable)

In 2024, WP6 updated the PANDASIA Knowledge Exchange Strategy Report (submitted as D6.1) along with the revised communication indicators (Appendix 1). SUPA 71 strongly believes that the knowledge exchange strategy should remain current and responsive to the communication needs of PANDASIA. The report has outlined the project's goals for communication, exploitation, and dissemination as follows:

Communication is defined as informing, creating awareness, and educating the intended audiences on project activities and scientific results as well as the societal challenges of dealing with future pandemics. PANDASIA's communication materials will feature the EU logo to highlight its foreign assistance efforts in Thailand. This branding not only symbolizes EU humanitarian support but also reflects its values globally. For local recipients, such as the Thai government, it may inspire ongoing support and funding for PANDASIA initiatives.

Dissemination is defined as sharing any output from our consortium with specific guidance, idea, publication, policy paper, algorithm and so on. The objectives on dissemination include: 1) Deliver open-source knowledge, 2) Develop global common goods and services with specific public health interventions, 3) Engage with wide array of stakeholders to apply, test out, and sustain our outputs, 4) Develop user friendly tools, apps, policy briefs, and technical guidance, 5) Network with various academic

Exploitation of results will ensure the sustainability of our work beyond the lifespan of the project by establishing sustainable partnerships with various academic, civil society organizations, international organization, and non-governmental organizations. WP6 will: 1) Utilize our results into generating national and regional policy guidance, 2) Develop educational material and curriculum, 3) Engage in capacity building activities, 4) Enhance national and regional application of the International Health Regulations (2005) and Pandemic prevention policies, 5) Develop innovative technologies including digital applications to enhance local preventive measures, 6) Develop global common goods and services with specific public health interventions.

The PANDASIA project is currently at the initial phase of data collection, although several scientific articles are in the pipeline to be submitted for publication in 2024, such as the policy and stakeholder mapping report, systematic review of public health messaging and community engagement in the Greater Mekong Subregion. Most results are expected to be ready for dissemination in 2025-2027. The WP6 will carry out community engagement activities for the dissemination of results to local authorities in year 4 and early year 5.

6. Appendix

6.1 Appendix 1: Certificate of renewal approval for human research by the Research Ethics Review Committee, Chulalongkorn University, July 2024

The Research Ethics Review Committee for Reseach Involving Human Research Participants, Group I, Chulalongkorn University

Chamchuri 1 Building, 2nd Floor, 254 Phayathai Road, Pathumwan, Bangkok 10330 Thailand

Telephone: 02-218-3202, 02-218-3049 Email: eccu@chula.ac.th

COA No. 168/67

Certificate of Approval

Study Title No. 660079 : PANDEMIC LITERACY AND VIRAL ZOONOTIC SPILLOVER RISK AT THE FRONTLINE OF DISEASE EMERGENCE IN SOUTHEAST ASIA TO IMPROVE PANDEMIC PREPAREDNESS: PHASE I

Principal Investigator : Prof. Dr. Alongkorn Amonsin

Place of Proposed Study/institution : Faculty of Veterinary Science, Chulalongkorn University

The Research Ethics Review Committee for Research Involving Human Research Participants, Group I, Chulalongkorn University, Thailand, has approved constituted in accordance with Belmont Report 1979, Declaration of Helsinki 2013, Council for International Organizations of Medical Sciences (CIOM) 2016, Standards of Research Ethics Committee (SREC) 2017, and National Policy and guidelines for Human Research 2015.

Signature Prida Tasanapradit

(Associate Prof. Prida Tasanapradit)

Chairman

Signature Raveenan Mingpakanee.

(Assistant Prof. Dr. Raveenan Mingpakanee)

Secretary

Date of Approval : 23 July 2024

Approval Expire date : 22 July 2025

The approval documents including:

1. Participant Information Sheet and Consent Form
2. Research proposal
3. Researcher
4. Research instruments / tools

Conditions

The approved investigator must comply with the following conditions:

1. It's unethical to collect data of research participants before the project has been approved by the committee.

2. The research/project activities must end on the approval expired date. To renew the approval, it can be applied one month prior to the expired date with submission of progress report.

3. Strictly conduct the research/project activities as written in the proposal.

4. Using only the documents that bearing the RECCU's seal of approval: research tools, information sheet, consent form, invitation letter for research participation (if applicable).

5. Report to the RECCU for any serious adverse events within 5 working days.

6. Report to the RECCU for any amendment of the research project prior to conduct the research activities.

7. Report to the RECCU for termination of the research project within 2 weeks with reasons.

8. Final report (AF 01-15) and abstract is required for a one year (or less) research/project and report within 30 days after the completion of the research/project.

9. Research project with several phases; approval will be approved phase by phase, progress report and relevant documents for the next phase must be submitted for review.

10. The committee reserves the right to site visit to follow up how the research project being conducted.

11. For external research proposal the dean or head of department oversees how the research being conducted.

The Research Ethics Review Committee for Research Involving Human Research Participants Group 1

Study Title No. 660079

Date of Approval 23 Jul 2024

Approval Expire date 22 Jul 2025

Digital Certificate

6.2 Appendix 2: Certificate of renewal approval for animal research by the Institutional Animal Care and Use Committee (IACUC), Chulalongkorn University, May 2024

Chulalongkorn University Animal Care and Use Committee

Certificate of Project Approval	☑ Original ☐ Renew
Animal Use Protocol **2331018**	
Protocol Title **Pandemic literacy and viral zoonotic spillover risk at the frontline of disease emergence in Southeast Asia to improve pandemic preparedness (PANDASIA)**	
Principal Investigator **Prof.Dr.Alongkorn Amonsin**	
Certification of Institutional Animal Care and Use Committee (IACUC) This project has been reviewed and approved by the IACUC in accordance with university regulations and policies governing the care and use of laboratory animals. The review has followed guidelines documented in Ethical Principles and Guidelines for the Use of Animals for Scientific Purposes edited by the National Research Council of Thailand.	
Date of Approval **May 24, 2024**	**Date of Expiration** **May 23, 2025**
Applicant Faculty/Institution **Faculty of Veterinary Science**	
Signature of Chairperson Chutamas Benjanirut	**Signature of Authorized Official**
Name and Title **Asst.Prof.Dr.Chutamas Benjanirut** **Chairman**	**Name and Title** **Assoc.Prof.Dr.Nuvee Prapasarakul** **Associate Dean for Research Innovation and Corporate Communication**
The official signing above certifies that the information provided on this form is correct. The institution assumes that investigators will take responsibility, and follow university regulations and policies for the care and use of animals. *This approval is subjected to assurance given in the animal use protocol and may be required for future investigations and reviews.*	

6.3 Appendix 1: Communication measures

The planning of communication measurement indicators will be revised/updated every four months, whenever necessary. The communication measures are formulated to support Communication, Exploitation and Dissemination. These will aid the project to check whether the identified target groups have been reached. Measurement indicators are major key performance indicators. They refer to a set of quantifiable measurements used to gauge our project's overall long-term performance.

#	Communication goals	Contributes to achieving PANDASIA overall goals	Key audiences	Messages	Communication measures and channels	Schedule	Measurement indicators (As of Dec 2024)	Status by December 30, 2024
1	Inform and educate the intended internal and external audience on project activities/ results as well as the societal challenges of preventing future pandemics. (*Communication*)	• Building synergies to prevent future pandemics. • Improving public awareness about relationships between humans, wildlife and their habitats, and zoonotic diseases. • Strengthening transdisciplinary collaboration and networking among consortium members. • Create and test a critical public health measure – a pandemic prevention and	• EU stakeholders • Thai stakeholders • Media & press • General public • Scientific Community/ academe • Civil society organizations • Industry • Donor community	**To be planned after obtaining all results from other WPs:** Continue to follow good hand and respiratory hygiene practices like regular handwashing and keep up to date with vaccination and be protected against viruses and bacteria causing diseases.	• PANDASIA Corporate Design	• 08/23	• Communication Strategy (Updated)	• Completed in Year 1 (2023)
					• PANDASIA Webpage with encrypted access and link to social media and website by partners	• 08/23	• PANDASIA webpage published online (06/23).	• Completed in Year 1 (2023)
					• Setup social media: X (formerly Twitter), Facebook, LinkedIn, and YouTube	• 02&06/23	• Social media platforms created & published	• Completed in Year 1 (2023)
					• PANDASIA Webpage Engagement	• 08/23-12/27	• 10,000 website page views per year.	• Achieved in 2023-2024, the website had a total of 21,588 page views. • Ongoing
					• Social media: X (formerly Twitter), Facebook, LinkedIn, and YouTube	• 08/23-12/27	• Social media /platform engagement target: ▪ 6,000 followers via social media platforms, including Twitter	• In 2024, achieved a grand total of 8,653 followers and connections

#	Communication goals	Contributes to achieving PANDASIA overall goals	Key audiences	Messages	Communication measures and channels	Schedule	Measurement indicators (As of Dec 2024)	Status by December 30, 2024
		preparedness literacy (3PL) intervention to reduce zoonotic transmission and pandemic risk.					(X), Facebook, LinkedIn, and YouTube. ▪ 100 postings and re-postings messages on LinkedIn. ▪ 100 postings and re-postings messages on X.	across all social media platforms. • 235 postings and re-postings messages on LinkedIn • 450 postings and re-postings messages on X.
					• Press releases about PANDASIA	• 06/23-12/27	• Press releases (PR): at least 1 PR per year	• Achieved 3 PRs in 2023-2024 • Ongoing
					• Network with media representatives (print/newspaper/ online, radio, TV)	• 12/24 -12/27	• Number of media contacts (print/newspaper/ online, radio, TV)	• Ongoing
					• PANDASIA glossary/dictionary	• 12/24 -12/27	• PANDASIA technical words popularized via posts in websites (scripts only in 2024): at least 1 PANDASIA Glossary.	• Ongoing

#	Communication goals	Contributes to achieving PANDASIA overall goals	Key audiences	Messages	Communication measures and channels	Schedule	Measurement indicators (As of Dec 2024)	Status by December 30, 2024
2	Ensure the use of project results in practice and decision-making to tackle the societal challenges of preventing future pandemics *(Exploitation)*	Improving public awareness about relationships between humans, wildlife and their habitats, and zoonotic diseases Create and test a critical public health measure – a pandemic prevention and preparedness literacy (3PL) intervention to reduce zoonotic transmission and pandemic risk.	• Thai local stakeholders • Civil society organizations • Public educational institutions • Industry	The health of people, animals, plants, and the environment are interconnected and must be addressed together:	• Approval/support /agreements of local stakeholders to support PANDASIA (in writing)	• 12/23	• Approval/support from local stakeholders	• Completed in Year 1 (2023)
					• Visits /presentations to local stakeholders	• To be determined in 2025 following the timeline of community engagement with WP1 and WP5.	• Number of town hall meetings (community engagement meetings, conferences, field meetings, meeting with local officers/community leaders conducted)	• Planning
					• Recommendations based on the policy/stakeholders mapping by WP1	• To be determined in 2025 following the timeline of a result from WP1 stakeholder mapping.	• Number of policy recommendations as a result from WP1 stakeholder mapping.	• Planning
					• Report on WP5 and WP6 collaboration on co-design for community response/solutions	• To be determined in 2025 following the timeline of community engagement with WP5.	• Co-designed messages	• Planning
					• Development of communication/training materials to be used by the community	• To be determined in 2025 following	• Communication materials/ training modules developed	• Planning

#	Communication goals	Contributes to achieving PANDASIA overall goals	Key audiences	Messages	Communication measures and channels	Schedule	Measurement indicators (As of Dec 2024)	Status by December 30, 2024
						the timeline of community engagement with WP5.	for health care workers	
					• Training and orientation of frontline workers on the use of communication materials to improve their communication skills and manner of messaging	• To be determined in 2025 following the timeline of community engagement with WP5.	• Number of training and orientation of frontline workers on the use of communication materials: at least 1 training.	• Planning
3	Translate project results into user-friendly formats or tools and make such results and products of evidence and make accessible to the scientific community, academia, industry, policy-making bodies, EU and Thai stakeholders. *(Dissemination)*	Improving public awareness about relationships between humans, wildlife and their habitats, and zoonotic diseases. Create and test a critical public health measure – a pandemic prevention and preparedness literacy (3PL) intervention to reduce zoonotic transmission and pandemic risk.	• Thai local stakeholders • Scientific community/ academe • Civil society organizations • Public educational institutions • Industry	• Professional, evidence based scientific "One Health" project supported by EU grant • Exchange of ideas and scientific results	• Scientific newsletter	• 08/23-12/27	• Number of newsletters: Annually/ Biannually	• Achieved 3 newsletters in 2023-2024 • Ongoing
					• Research briefs/ policy papers collaboration with WP1 and WP5	• To be determined in 2025 following the timeline of community engagement with WP5	• Number of research briefs/ policy papers: at least 1 research briefs/ policy papers	• Planning
					• Conferences	• 08/23-12/27	• Number of conferences attended and presented the PANDASIA's works and activities by PANDASIA consortium partners (onsite and online): Annually/ Biannually	• 10 conferences or major meetings were attended across the work packages in 2024.

#	Communication goals	Contributes to achieving PANDASIA overall goals	Key audiences	Messages	Communication measures and channels	Schedule	Measurement indicators (As of Dec 2024)	Status by December 30, 2024
					• Stakeholder meetings	• 10/23	• Number of Workshops conducted: at least 2 meetings	• Completed 2 meetings of stakeholder workshops in Year 1 (2023)
					• Conduct of research seminar/paper presentation/ research dissemination forum	• To be determined in 2025 following the timeline of community engagement with WP5	• Number of scientific seminars held/assisted	• Planning
					• Research paper/manuscript for publication	• 12/27	• Number of publications: at least 1 publication	• Planning